An Arnold-Chiari malformation (also referred to as Chiari malformation) is a congenital defect in which part of all of the cerebellum is displaced downwardly. The different forms of Chiari malformations are classified as follows:

Class	Structure herniated
Type I	Cerebellar tonsils
Type II	Cerebellar vermis, fourth ventricle, and lower brainstem
Type III	Entire cerebellum is displaced caudally
Type IV	Cerebellar hypoplasia

The signs and symptoms of a Chiari I malformation can present at any age and generally include occipital headache and pain in the shoulders and arms. Visual disturbances, vertigo, and ataxia are very common symptoms and symptoms of syringomyelia often occur in patients with a Chiari I malformation.

A Chiari II malformation typically present in infancy and are associated with hydrocephalus and brainstem and cranial nerve dysfunction.

Surgical treatment is aimed at decompressing the cerebellum, enlarging the foramen magnum, and freeing any adhesions. This is often performed via cervical laminectomies and craniectomy surgery. Key anesthetic considerations are preventing increases in intracranial pressure and managing blood loss, especially in the case of Chiari II malformations.

Medullary cord compression can occur, particularly with extension of the neck and patients may present with symptoms such as vocal cord paralysis, swallowing dysfunction, and chronic pulmonary aspiration.

Cote CJ, Lerman J, & Anderson BJ. A Practice of Anesthesia for Infants and Children. 5th Ed. Philadelphia: Elsevier Saunders; 2013: 530.
Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 609.

Copyright 2015 Prodigy Anesthesia, Inc. It is unlawful to copy, distribute, and/or sell this material.